

Universal Pre-training for Images

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Abstract

This thesis investigates whether Vision Transformers can acquire transferable visual representations from image-like data generated by iterated random computation (IRC), without using real images during pre-training. Universal pre-training motivates this question: independent random noise provides little learnable structure, whereas computation can transform randomness into structured dependencies. We implement an IRC image generator that combines random spatial operations and residual convolutional updates in a persistent buffer, and use its output to pre-train a ViT-Tiny encoder with a masked-autoencoder objective for 100 million image presentations. We evaluate frozen representations on nine validation datasets (including three synthetic diagnostics) and four test datasets, and also report reconstruction errors without downstream adaptation. The model reaches a mean classification accuracy of 56.75% on the validation group and 32.60% on the test group, compared with 29.39% and 17.02% for its untrained encoder. In separate fine-tuning experiments, IRC-MAE pre-training improves final accuracy over training from scratch by 1.20 percentage points on ImageNet-100 and 5.23 points on a 10% ImageNet-1K subset. An I-JEPA experiment provides a comparison with prediction in representation space. The results show transfer from synthetic pre-training to real-image tasks under the tested configurations; repeated seeds are needed to quantify the variability of these gains.

1 Introduction

Larger models, trained on larger datasets, often learn representations that transfer better, converge faster, and generalize more robustly to downstream tasks [1, 2, 3]. In computer vision, this has led to a standard recipe: first pre-train an image encoder on a large generic dataset, and then fine-tune it on a smaller target dataset [4, 5, 2]. This pipeline has been central to the success of modern visual representation learning, especially for Vision Transformers (ViTs), which process images as sequences of patch tokens and inherit much of their design from the Transformer architecture used in sequence modeling [6, 3]. This approach requires access to a large collection of real images.

This assumption is not always realistic. In many domains, data is expensive, restricted, private, or simply unavailable at the required scale. One domain in which this difficulty is particularly acute is medical imaging. Medical images are often protected by privacy regulations, collected under heterogeneous acquisition protocols, and tied to sensitive patient information [7]. Even when images can be accessed, labels may require expert annotation, and for rare diseases the number of positive examples may be intrinsically small [7]. Moreover, medical images can vary across hospitals, scanners, acquisition settings, and patient populations, making small datasets even harder to use reliably [8]. These constraints motivate a central question for data-efficient vision: can we reduce the dependence of visual pre-training on large real-image datasets?

Self-supervised learning removes one important bottleneck, namely the need for manual labels, but it does not remove the need for images [9, 10, 11]. Most established self-supervised vision methods are trained on large collections of real images [12, 13]. Thus, the scarcity problem concerns not only labels but also data availability. If the target domain is private, specialized, or rare, then even unlabeled data may be limited. In such settings, a useful initialization can be as valuable as an additional dataset: it may reduce the amount of real domain data required to reach a given level of performance [14, 8].

The difficulty becomes clearer when considering the size of image space. An RGB image of height H and width W , with 8-bit color channels, belongs to a space of

$$256^{H \cdot W \cdot 3}$$

possible arrays. Even for a relatively standard ViT input resolution of 224×224 , this is approximately $10^{362,500}$ possible images. For a 1024×1024 image, which is still modest compared with some high-resolution medical and scientific images, the number is approximately $10^{7,575,700}$. By comparison, the number of atoms in the observable universe is often estimated to be on the order of 10^{80} [15]. Exhaustive coverage of image space is infeasible.

Yet natural images occupy only a tiny and highly structured subset of this space [16]. An image sampled uniformly at random from this space is therefore almost certain to be entirely unstructured. Useful images, in contrast, contain regularities: neighboring pixels are often correlated, edges form contours, contours enclose regions, textures repeat, objects have boundaries, shapes compose into scenes, and distant patches can depend on one another [17, 18, 19]. The aim is to learn these regularities without collecting real images for pre-training.

If we design the generator ourselves, where does the useful training signal come from? In information-theoretic terms, computation cannot create new information from nothing [20, 21]. However, useful training data is not valuable merely because it contains many bits. Pure random noise has maximal unpredictability, but no learnable structure. What matters for pre-training is the combination of variation and regularity: randomness provides diversity, while computation can turn that randomness into dependencies, repetitions, hierarchy, and constraints [21].

A useful intuition is to compare two kinds of randomness. In the first case, we sample every pixel of an image independently. The result is similar to television static: it contains many bits, but almost no exploitable structure. In the second case, we first sample random pixels and then pass them through a randomly chosen computational process. This process might blur local neighborhoods, copy patches, impose symmetries, threshold intensities, warp coordinates, apply filters, or repeatedly feed intermediate outputs back into itself. For fixed transformation parameters, deterministic computation cannot increase the entropy of a discrete input, and it does not supply real-world semantics [20, 21]. However, it can transform independent randomness into structured randomness. The resulting images may not depict real objects, but they can contain edges, regions, textures, repeated patterns, spatial continuity, and dependencies between distant patches. These dependencies provide a prediction task even when the images have no semantic labels.

This perspective is closely related to *universal pre-training*. Instead of first collecting data from the target domain, universal pre-training asks whether a model can be pre-trained on data generated before the downstream task is known. In the sequence setting, *Universal pre-training by iterated random computation* proposes to generate structured random data by passing randomness through random computational processes, and to use this data to pre-train a model that can later adapt to many different tasks [21]. The most important idea is not that the generated data contains the semantics of future tasks. Rather, the hope is that many real-world data sources are themselves produced by computational or physical processes, and therefore share broad structural properties.

Generator-specific patterns should average out across many instances, whereas higher-level patterns that recur across instances should be reinforced in the training distribution [21].

The same view naturally suggests a data–computation tradeoff. When real data is abundant, a model can learn structure directly from examples. When real data is scarce, private, or expensive, we may instead spend computation to generate structured pre-training signals [21]. Such pre-training can then be amortized over many downstream tasks: the expensive synthetic pre-training stage is performed once, while the resulting initialization can be reused whenever real target data is limited. This is especially attractive for domains such as medical imaging, where obtaining an additional large dataset may be harder than spending additional computation.

Vision Transformers provide a natural testbed for this idea. A ViT converts an image into a sequence of patches and processes these patches with a Transformer encoder [6, 3]. This patch-sequence formulation creates a bridge between sequence learning and image learning: the input is visual, but the architecture reasons over tokens and their relations. At the same time, ViTs contain fewer image-specific inductive biases than convolutional networks. Locality, translation equivariance, and two-dimensional neighborhood structure are not built into every layer in the same way as in CNNs; much of the spatial structure must be learned from data [3]. This makes ViTs powerful, but also data-hungry: large ViTs tend to benefit most from large-scale pre-training, while smaller or synthetic pre-training regimes require careful design to compensate for the weaker built-in visual priors [3, 23, 24].

There is already evidence that useful visual representations need not come exclusively from natural photographs. Prior work has shown that models can learn from procedural noise processes, provided that the generated images contain suitable structural properties such as naturalistic statistics and sufficient diversity [16]. Formula-driven supervised learning and fractal-based datasets have also shown that visual encoders can be pre-trained on synthetic images generated from mathematical rules rather than real image collections [25, 26]. In particular, one-instance fractal datasets suggest that even a very small number of synthesized images can provide useful ViT pre-training when combined with appropriate augmentation [24]. More broadly, generated image–caption pairs have also been shown to support strong visual representation learning without directly training on real images [27].

Recent work pushes this idea even further. Procedural warm-up for Vision Transformers shows that non-visual symbolic data, generated from formal grammars and used before image-based training, can improve convergence and downstream performance [28]. This suggests that some helpful priors for vision may be computational rather than purely visual. A model may benefit from learning generic mechanisms such as dependency tracking, hierarchy, composition, and long-range interaction, even before it is exposed to pixels.

Synthetic images, fractals, structured noise, generated image–caption pairs, and symbolic procedural data have all been explored as alternatives or complements to real visual data [16, 25, 24, 27, 28]. Separately, universal pre-training has been studied for token sequences and explicitly proposed as a direction that could, in principle, extend to computer vision [21]. However, it remains unclear how universal random computation should be adapted directly to image-based ViT pre-training. In particular, the vision setting raises new questions: what kind of generated image-like structure is useful, how should randomness and spatial regularity be balanced, and which visual priors can be learned before the model sees real images?

Therefore, this thesis asks the following research question: *Can Vision Transformers acquire transferable visual representations from image-like data generated by iterated random computation, without using real images during pre-training?* The goal is not to replace domain-specific data entirely. Real data remains necessary for evaluation, adaptation, and learning task-specific semantics. Instead, the goal is to reduce the amount of real data required by starting from a more useful

initialization. Medical imaging motivates this question, but the experiments here evaluate general visual transfer rather than medical applications.

2 Related Work

Early visual pre-training methods trained on large labeled image datasets and transferred the learned representations to smaller tasks. ImageNet played a central role in this story, not only as a benchmark, but also as a reusable source of visual features [4, 5]. Later, larger datasets such as JFT and large-scale transfer methods showed that increasing the amount of pre-training data could further improve downstream performance [1, 2].

Vision Transformers made this dependence on data especially visible. By representing images as sequences of patches, ViTs brought the Transformer architecture from language into vision [6, 3]. This made the architecture simple and scalable, but also reduced some of the image-specific inductive biases that convolutional networks have by design. Locality, translation equivariance, and two-dimensional neighborhood structure are not built into ViTs in the same strong way as in CNNs; instead, many spatial relations must be learned from data [3]. This helps explain why large-scale pre-training is so important for ViTs, and why data-efficient ViT training has become an important research direction [23, 24].

A first step away from labeled data is self-supervised learning. Methods such as SimCLR, MoCo, DINO, MAE, and iBOT show that strong visual representations can be learned without manual labels, by designing learning objectives from the images themselves [29, 30, 9, 10, 11]. More recent foundation models such as DINOv2 and DINOv3 scale this idea further and demonstrate that self-supervised image encoders can become broadly useful across classification, dense prediction, and other vision tasks [12, 13]. However, these methods still depend on large collections of real images. They remove the label bottleneck, but not the image bottleneck. For domains where even unlabeled data is scarce or private, this distinction matters.

Another line of work asks whether we can use real data more efficiently. DeiT showed that training choices such as distillation and augmentation can make ViTs more data-efficient [23]. More radically, DORA investigates whether a strong image encoder can be learned from a single long unlabeled video, rather than from a large collection of independent images [31]. This result is important for the present thesis because it suggests that the organization of data can matter as much as its raw size. A continuous video contains temporal coherence, object persistence, viewpoint changes, and natural transformations. These regularities offer a source of visual variation within a single recording.

A more direct challenge to real-image pre-training comes from synthetic and procedural visual data. FractalDB and formula-driven supervised learning show that models can be pre-trained on images generated from mathematical rules rather than collected from cameras [25]. Follow-up work on one-instance fractal databases shows that even a very small number of synthesized images can be useful for ViT pre-training when combined with suitable augmentation [24].

This idea is also developed in work on learning from noise. Baradad et al. study several procedural noise processes and show that visual representations can be learned from generated images that are not natural photographs [16]. Their main message is especially relevant here: the best synthetic data is not necessarily the most realistic data. Instead, useful generated data should balance diversity with structural properties similar to those found in natural images. Procedural Image Programs extend this direction by using richer procedural generators as a source of visual pre-training data [32].

A different approach is to use generative models as data sources. SynCLR, for example, uses

language models to synthesize captions and text-to-image models to synthesize images, then learns visual representations contrastively, treating images generated from the same caption as positive pairs [27]. This shows that a visual encoder can be trained without directly using a real-image dataset as its pre-training set. However, the source of structure in this case still comes from powerful generative models that were themselves trained on large real-world corpora. SynCLR therefore differs from the setting considered in this thesis. Here, the goal is not to distill visual knowledge from already-trained generative models, but to ask whether useful structure can arise from random computation itself.

The most closely related conceptual starting point is universal pre-training. Universal pre-training studies whether a model can be trained before the downstream data is known, using synthetic data generated from random computational processes [21]. Grau-Moya et al. take a related approach: they generate meta-training data with Universal Turing Machines and train neural networks to learn general prediction strategies [22]. The key idea is that pure randomness is not useful, but randomness passed through computation can become structured. Such data may contain dependencies, repetition, hierarchy, and regularities that are not specific to one dataset. If many real-world data sources are produced by computational or physical processes, then a model trained on many random computational sources may learn patterns that are broadly reusable. Universal pre-training therefore frames pre-training as a data-computation tradeoff: when real data is limited, one may spend computation to generate useful structure [21].

Finally, Procedural Warm-Up for Vision Transformers connects this idea directly to vision models. Instead of using images, it pre-trains ViTs on symbolic sequences generated by formal grammars before standard image training [28]. Shinnick et al. find that non-visual procedural data can improve convergence and downstream performance. This suggests that some useful priors for vision may be computational rather than purely visual: hierarchy, dependency tracking, composition, and long-range interaction may help a Transformer before it ever sees pixels.

Prior work ranges from labeled and unlabeled real images to synthetic images and non-visual procedural data. This thesis studies whether iterated random computation can directly generate useful image-like pre-training data for Vision Transformers before the model sees any real images.

3 Approximating Universal Pre-training for Images

The previous sections motivate the main idea of this thesis: useful pre-training data does not necessarily have to be real, but it should contain learnable structure. Universal pre-training gives a general version of this idea. In principle, one would like to sample data from a very broad distribution over computable sources, so that a model can learn patterns that are not tied to one specific dataset or task [21]. In practice, however, the true universal distribution cannot be sampled directly. We cannot enumerate all programs, we cannot run them for unbounded time, and we cannot store an infinite stream of generated data.

The method used in this thesis is therefore a practical approximation. It does not claim to sample from the true universal distribution. Instead, it tries to capture one useful principle behind it: start from randomness, repeatedly apply randomly sampled computations, and train a model on the structure that appears. In the image setting, this means maintaining a buffer of random image-like tensors and repeatedly applying spatial operations and randomly initialized convolutional networks. The generated images are not designed to look like natural photographs. Their role is more basic: they should be structured enough for a Vision Transformer to learn spatial and computational regularities. This thesis evaluates this construction empirically; a formal analysis of its relation to the universal distribution is left to future work.

3.1 From Random Computation to Image-like Data

Universal pre-training was originally studied in a sequence setting, where random bits are passed through random computations to produce structured token sequences [21]. Images are different from text, but Vision Transformers make the connection natural. A ViT converts an image into a sequence of patch tokens and processes those tokens with a Transformer encoder [6, 3]. Therefore, the vision version of the problem can be phrased as follows: can we generate image-like tensors whose patch sequences contain useful structure?

Let an image be represented as

$$x \in [0, 1]^{3 \times R \times R},$$

where R is the image resolution. In all our experiments, $R = 224$. The generator starts from random images in this space and repeatedly applies random image computations. The family of computations used here is intentionally simple: small randomly initialized convolutional networks together with crop-and-resize, blur, and crossover operations. This is not as general as all possible programs, but it is a reasonable first step for images. Convolutions preserve the spatial layout of the image, create local dependencies, and can be applied repeatedly to increase the effective computational depth.

This choice also keeps the method conceptually close to the motivation from the introduction. Pure pixel noise contains many bits, but no predictable organization. After random convolutional processing, the image may contain smooth regions, color correlations, edges, textures, repeated motifs, and larger spatial dependencies. These are not semantic objects, but they are still forms of structure.

3.2 The IRC Image Generator

We call this method the IRC image generator. It keeps a persistent buffer

$$\mathcal{B}^{(t)} = \{x_1^{(t)}, x_2^{(t)}, \dots, x_B^{(t)}\},$$

where each $x_i^{(t)}$ is a generated image at training step t . In the final configuration, the buffer contains $B = 40,000$ RGB images of size 224×224 . Half of the buffer is initialized from independent uniform pixel noise, and the other half contains random solid-color images. Pixel values are stored as integers from 0 to 255 and divided by 255 before computation. Every image starts at depth zero.

Before sampling a training batch, the generator first resets 0.75% of the buffer, or 300 images. Half of these entries receive new random solid colors and half receive independent pixel noise. Their depths are reset to zero. It then selects 2,000 distinct buffer entries to update and independently selects 2,000 partner entries. The two sets may overlap. The updates are processed in chunks of 200 images, with a newly sampled spatial program and random convolutional network for each chunk.

A spatial program is a short sequence of crop-and-resize, blur, identity, and crossover operations. At each position, identity has probability $1/2$, and each of the other three operations has probability $1/6$. After an operation, the program stops with probability $1/2$, with a maximum length of 100 operations. The expected length is therefore approximately two operations. Before the spatial program is applied, each selected source image is also replaced with probability 0.05 by fresh pixel noise or a fresh solid color, with equal probability. This replacement resets its source depth.

Crop-and-resize selects a square crop with a side-length scale sampled uniformly between 0.5 and 0.9, chooses a position that keeps the crop inside the image, and resizes it to the original resolution using bilinear interpolation. Blur uses a uniform averaging kernel with side length 3, 5, or 9, sampled with equal probability, and reflection padding. Identity leaves the image unchanged.

Crossover combines a source image x and a partner image y . A random field with a spatial grid of 2×2 , 4×4 , or 8×8 is sampled and upsampled to the image resolution using bicubic interpolation. The field is thresholded at a value sampled uniformly between 0.35 and 0.65, producing a binary mask m . The new image is

$$x' = m \odot x + (1 - m) \odot y.$$

The same mask is used for all three color channels. This operation copies spatial regions from the two parents rather than averaging their pixels.

After the spatial program, a randomly initialized residual convolutional network is applied. It contains two 3×3 convolutions, mapping 3 channels to 16 and then back to 3, with a LeakyReLU activation of slope 0.2 between them. Both convolutions use reflection padding, zero biases, and Kaiming-normal weight initialization. The update is

$$x_{\text{new}} = \text{clip} \left(x' + 0.1 \tanh \left(g_{\phi}(x' - 0.5) \right), 0, 1 \right).$$

The residual connection retains the current image while adding a bounded random change. The generator weights are never trained. The result is rounded to the nearest multiple of $1/255$ before being stored in the buffer.

3.3 Why the Buffer Matters

Images remain in the buffer between calls, so repeated updates can accumulate the effects of different random programs and networks. Each buffer position has probability $2,000/40,000 = 0.05$ of being selected for an update at a given step. Without persistence, each image would receive only the computation performed during one call.

The generator records a depth for every image. If the sampled program contains crossover, the new depth is the maximum of the source and partner depths plus one. Otherwise, it is the source depth plus one. This is a record of the generator’s computational ancestry, rather than a direct measurement of visual complexity or learning difficulty. In particular, identity, smoothing, and crossover masks that select a single parent need not make an image harder to predict, even when its recorded depth increases.

The training batch is sampled without replacement from entries whose depth is at least the 20,000th-largest depth in the buffer. Ties are retained, so the eligible set can contain more than 20,000 images. This favors images that have accumulated computation while the resets maintain a supply of fresh sources. The reset rate and residual update strength remain constant throughout training. No Fourier textures are injected.

An idealized description is a mixture over computation depths,

$$q_{\text{mix}}(x) = \sum_{d \geq 0} \pi_d q_d(x).$$

Here q_d denotes the image distribution at recorded depth d , and π_d is its sampling weight. The weights depend on the evolving buffer and the depth-based sampling rule. This construction is a restricted empirical approximation motivated by universal pre-training; it is not an implementation of the universal distribution itself.

4 Training

4.1 Target Model and Pre-training Objective

The target model is a Vision Transformer trained with a masked-autoencoder objective. The final run uses a ViT-Tiny/16 encoder with 224×224 input images and 16×16 patches. This gives

$$N = \left(\frac{224}{16}\right)^2 = 196$$

image patches. Each patch is projected into a 192-dimensional embedding. The encoder has depth 12, 3 attention heads, MLP ratio 4.0, and no dropout. A learnable class token and fixed two-dimensional sinusoidal positional embeddings are used.

Since the IRC images have no semantic labels, the pre-training objective cannot rely on class supervision. Instead, this thesis uses masked image modeling, following the general idea of Masked Autoencoders [10]. For each generated image, 75% of the patches are masked. The encoder receives only the visible patches, and a small decoder reconstructs the missing RGB patches. The decoder has embedding dimension 128, depth 2, and 4 attention heads. Patch-wise target normalization is disabled; the targets are raw RGB values in $[0, 1]$.

Let $\mathcal{M}$ be the set of masked patches. The pre-training loss is

$$\mathcal{L}_{\text{MAE}} = \frac{1}{|\mathcal{M}|} \sum_{j \in \mathcal{M}} \frac{1}{D} \|\hat{p}_j - p_j\|_2^2,$$

where p_j is the original flattened RGB patch, $\hat{p}_j$ is the reconstructed patch, and $D = 16 \cdot 16 \cdot 3$ is the patch dimensionality.

4.2 Training Process

The pre-training loop alternates between generating images and updating the target model. One training step proceeds as follows:

1. Reset 300 buffer entries to fresh independent noise or solid colors, with equal proportions.
2. Update 2,000 selected entries using random spatial programs and residual convolutional networks, as described in Section 3.2.
3. Sample a batch of 4,000 images without replacement from the eligible depth range.
4. Mask 75% of the image patches and reconstruct their raw RGB values with the MAE.
5. Update the target model with AdamW. The generator weights are never trained.

The training budget is 100 million image presentations, corresponding to 25,000 optimizer steps. This counts sampled images, including repeated samples from the persistent buffer, rather than distinct images. The initial model and every subsequent 4 million-image checkpoint are retained. AdamW uses $\beta = (0.9, 0.95)$ and weight decay 0.05, with biases and one-dimensional parameters excluded from weight decay. The base learning rate is 1.5×10^{-4} , scaled by batch size over 256 to a peak of 2.34375×10^{-3} . A linear warm-up covers the first 5% of training, followed by cosine decay to zero.

The primary IRC-MAE experiment uses the implementation fixed before training.¹ Pre-training used one H100 GPU on Snellius; checkpoint evaluation ran on DAS-6.

¹<https://github.com/VladMihaiNechita/Master-Thesis>, revision d6661dd.

Component	Configuration
Target model	ViT-Tiny/16, 224×224 RGB input
Encoder	dimension 192, depth 12, heads 3, MLP ratio 4
MAE decoder	dimension 128, depth 2, heads 4
Mask ratio and target	0.75; raw pixels without patch normalization
Buffer	40,000 images; half pixel noise, half solid colors initially
Resets	300 entries per step; equal noise/color mixture
Updated entries	2,000 per step, in chunks of 200
Source replacement	probability 0.05 before the spatial program
Spatial operations	identity, crop-and-resize, blur, crossover
Residual CNN	$3 \rightarrow 16 \rightarrow 3$ channels, 3×3 kernels, strength 0.1
Training sampling	depth at least the 20,000th-largest buffer depth
Quantization	256 levels per channel
Training budget	100 million image presentations; batch size 4,000
Optimizer	AdamW; $\beta = (0.9, 0.95)$; weight decay 0.05
Learning rate	peak 2.34375×10^{-3} ; 5% warm-up, cosine decay
Checkpoints	initialization and every 4 million images
Random seed	1729

Table 1: IRC-MAE pre-training configuration for the primary experiment.

5 Evaluation

5.1 Downstream Evaluation

After pre-training, we discard the MAE decoder for classification and freeze the encoder. Each downstream image is represented by the mean of the final patch-token representations, excluding the class token:

$$h = \frac{1}{N} \sum_{j=1}^N h_j.$$

The ViT-Tiny encoder therefore produces a 192-dimensional feature vector. The classifier consists of non-affine batch normalization followed by a linear layer. We train it for 100 epochs using SGD, with batch size 256, momentum 0.9, zero weight decay, and peak learning rate 0.1. The learning rate warms up from 0.01 over 10 epochs and then follows cosine decay over the remaining 90 epochs. This protocol measures the quality of the representation without allowing the encoder to adapt to the downstream task.

In addition to classification accuracy, we report reconstruction error without downstream adaptation. Classification tests whether the encoder has learned representations useful for recognizing categories. Reconstruction tests whether the MAE can recover missing patch content, including colors, edges, contrast, and fine texture, on distributions it did not encounter during pre-training.

Following the masked-autoencoder objective, we compute mean squared error on masked raw RGB patches [10]. For target patch $p_j \in \mathbb{R}^D$, prediction $\hat{p}_j$, and $D = 16 \cdot 16 \cdot 3$,

$$\text{MSE}_{\text{raw}}(p_j, \hat{p}_j) = \frac{1}{D} \sum_{d=1}^D (\hat{p}_{j,d} - p_{j,d})^2.$$

The mask ratio is 0.75, and the loss is averaged over the complete evaluation split. The same seed is set for each dataset evaluation. We report raw-pixel MSE without patch normalization.

Lower raw-pixel MSE means that reconstructed RGB values are closer to their targets. However, a model may obtain a low error by predicting smooth average colors while missing sharper details. Reconstruction is therefore not a substitute for downstream accuracy. The two measurements should be read together: classification measures the transfer of the representation, while reconstruction measures how well the pre-training objective transfers to the pixel structure of other images.

5.2 Datasets and Splits

The downstream datasets are used to evaluate saved checkpoints, not to update the encoder during IRC pre-training. For linear probing, the encoder remains frozen and only the classifier is trained on labeled downstream images.

The evaluation suite is split into two groups. The first group is used as a *validation* group during development. These datasets are useful for comparing generator choices, inspecting training behavior, and understanding which kinds of visual structure are learned. The second group is the *test* group. For the primary experiment, the implementation and 100 million-image endpoint are fixed before its evaluations. Every checkpoint is evaluated on both groups to show progress, but test results are not used to select a checkpoint or change the generator. These test datasets were also available during earlier research, so this is not a previously unseen benchmark suite.

The suite covers object and fine-grained recognition, texture, remote sensing, and scene classification. Three synthetic tasks provide controlled diagnostics for shape, orientation, and texture. ImageNet-100 and the 10% ImageNet-1K training subset are used separately for fine-tuning.

All images are evaluated at resolution 224×224 , matching the ViT input resolution used during pre-training. For small low-resolution datasets, the whole image is resized directly to 224×224 . For larger natural-image datasets, the shorter side is first resized to 256, followed by a center crop of size 224×224 . This keeps evaluation deterministic and avoids introducing extra augmentation during evaluation.

Dataset	Type	Preprocessing	Purpose
CIFAR-10 [33]	Real, val	Resize 32×32 to 224×224	Basic object recognition.
Tiny-ImageNet [34]	Real, val	Resize 64×64 to 224×224	Object recognition across 200 classes.
Food-101 [35]	Real, val	Resize short side to 256, center crop 224	Fine-grained food recognition.
Oxford-IIIT Pets [36]	Real, val	Resize short side to 256, center crop 224	Fine-grained animal recognition.
DTD [37]	Real, val	Resize short side to 256, center crop 224	Recognition of natural textures.
EuroSAT RGB [38]	Real, val	Resize 64×64 to 224×224	Land-use and land-cover classification.
ShapeColorObjects	Synthetic, val	Generated at 224×224	Shape recognition with varying colors and backgrounds.
LineFieldOrientation	Synthetic, val	Rendered at high resolution, then antialiased to 224×224	Orientation and spatial-frequency discrimination.
TextureMosaic	Synthetic, val	Generated at 224×224	Procedural texture-family recognition.

Table 2: Validation datasets used during development and analysis. These datasets are used after pre-training with a frozen encoder and a linear classifier.

Dataset	Type	Preprocessing	Purpose
CIFAR-100 [33]	Real, test	Resize 32×32 to 224×224	Object recognition across 100 classes.
Stanford Cars [39]	Real, test	Resize short side to 256, center crop 224	Fine-grained car recognition.
RESISC45 [40]	Real, test	Resize 256×256 to 224×224	Remote-sensing scene recognition.
SUN397 [41]	Real, test	Resize short side to 256, center crop 224	Scene and place recognition.

Table 3: Four test datasets used for the fixed-endpoint comparison. SUN397 uses its first standard partition.

For the real datasets, we train the classifier on the training split and evaluate it on the corresponding evaluation or test split. The same data partitions are used for every checkpoint; SUN397 uses its first standard partition. The validation/test grouping refers to different downstream datasets, not to partitions of the generated pre-training buffer.

The synthetic validation datasets are generated only for evaluation. They are not used during IRC pre-training. Each contains approximately 50,000 training images and 10,000 test images; the exact sizes are rounded down when necessary so that every class has the same number of samples. The evaluation images are generated separately from the pre-training buffer, with balanced classes.

Figure 1 shows representative examples from the three synthetic validation datasets. ShapeColorObjects varies the foreground color, background, and distractor objects while keeping the main shape as the label. The other two datasets combine several regions in each image, so the labeled line or texture pattern appears alongside competing patterns rather than in isolation.

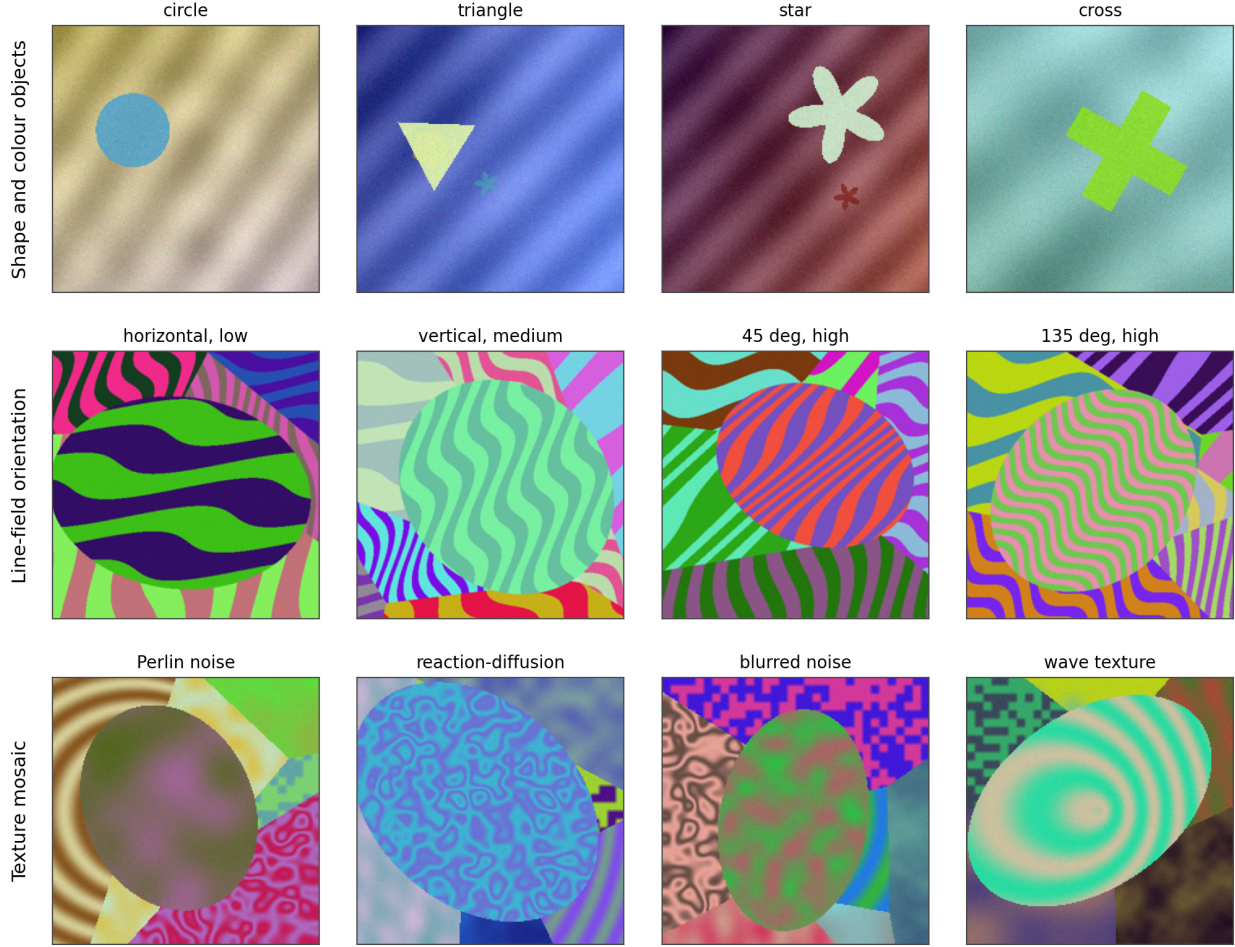


Figure 1: Representative test samples generated by the evaluation code. Each row corresponds to one synthetic validation dataset, and the title above each image gives its class label.

ShapeColorObjects. This dataset contains 8 classes:

$$\{\text{circle, square, triangle, ellipse, star, ring, cross, polygon}\}.$$

Each image contains one main foreground object on a random background. The main object can be a circle, square, triangle, ellipse, star, ring, cross, or polygon. Its position, scale, rotation, color, background texture, and partial occlusion are randomized. The classification label is the main object shape.

LineFieldOrientation. This dataset contains 12 classes, formed by 4 orientations and 3 frequencies:

$$4 \times 3 = 12.$$

The orientations are horizontal, vertical, diagonal 45° , and diagonal 135° . The frequencies are low, medium, and high. Each image is made of several irregular regions, where each region contains a line field with a specific orientation and frequency. The classification label is the dominant orientation/frequency class in the image.

TextureMosaic. This dataset contains 8 procedural texture classes:

{perlin_noise, fourier_texture, reaction_diffusion,
cellular_automata, fractal_texture, blurred_noise,
dot_texture, wave_texture}.

Each image is a mosaic of irregular regions, and each region is filled with one procedural texture family. The classification label is the dominant texture family in the image. This dataset is useful because it directly tests whether IRC pre-training helps the model represent texture-like and region-level structure.

Generator samples. Figure 2 shows samples from the generator at initialization, after 4 million images, and after 100 million images. The initial buffer mixes independent pixel noise and random solid colors. Repeated spatial operations and residual convolutions subsequently transform these sources into regions, boundaries, color transitions, and textures. Crossover can combine structures from different buffer entries, while blur and crop-and-resize change their spatial scale. The images remain abstract and do not explicitly depict natural-image categories.

Each panel uses the first 16 images in the batch logged at that checkpoint, without selecting examples for their appearance. Checkpoint batches follow the generator’s depth-based sampling rule, so they represent the images eligible for training rather than a uniform sample of the entire buffer. The panels show different images, not tracked trajectories. These examples illustrate the distribution but do not measure complexity as a function of depth.

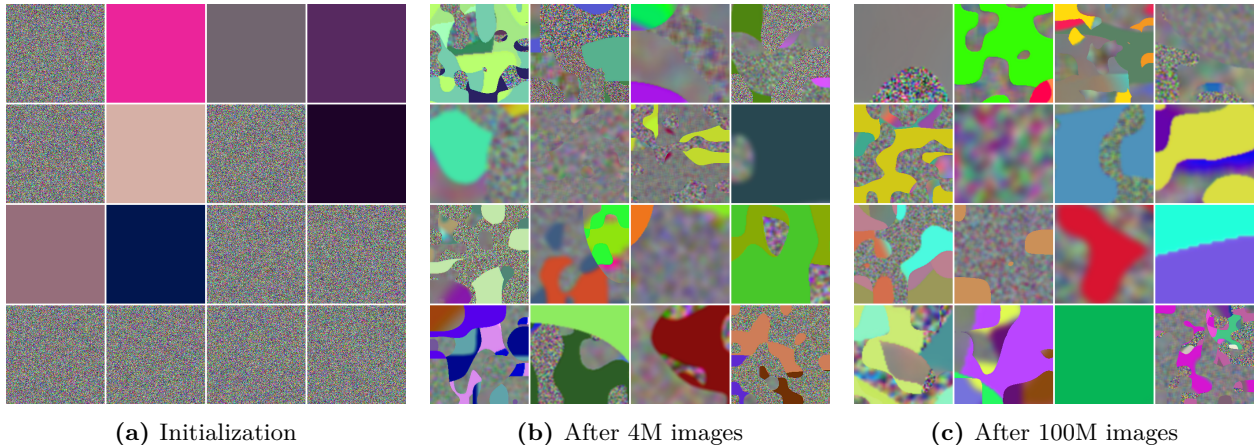


Figure 2: Samples from the training-eligible generator distribution at three fixed checkpoints. Each panel contains the first 16 logged images at that checkpoint.

6 Experimental Results

6.1 Baselines and Comparison Protocol

We compare the primary IRC-MAE model before and after pre-training. The no-pre-training reference is the same ViT-Tiny encoder before its first optimizer update. Its weights remain fixed during linear probing, just as for each pre-trained checkpoint. This provides a direct comparison under the same architecture, dataset preprocessing, probe training, and evaluation implementation.

We also evaluate the retained 100 million-image checkpoint from the earlier IRC-MAE experiment with the same evaluator, and refer to it as the historical IRC-MAE run. Archived validation curves additionally compare IRC with Fourier textures and two sampling variants of a real-video source. The video uses frames from a single walking recording, with continual or stratified sampling, following the single-video setting of DORA [31]. These archived source comparisons have compatible recorded model settings, but their exact source code and runtime conditions are not fully verified; they are not controlled ablations of individual generator operations.

6.2 Frozen Representations

Figures 3 and 4 show the primary run’s checkpoint trajectories. Table 4 reports the fixed final endpoint, without selecting a different checkpoint for each dataset. Group means weight each dataset equally; in particular, the nine-dataset validation mean includes three synthetic diagnostics and should not be read as a mean over natural-image tasks alone. IRC-MAE improves over its untrained encoder on all 13 datasets. The mean gain is 27.37 percentage points on the validation group and 15.59 points on the test group. Across only the six real validation datasets, the mean changes from 28.18% to 43.89%.

Dataset	Untrained	IRC-MAE	Gain (pp)	Historical IRC
<i>Validation datasets</i>				
CIFAR-10	46.43	63.06	+16.63	62.88
DTD	14.79	34.79	+20.00	33.46
EuroSAT	71.06	90.46	+19.41	90.70
Food-101	13.14	26.59	+13.45	26.26
Oxford-IIIT Pets	12.05	23.30	+11.26	22.38
Tiny-ImageNet	11.63	25.14	+13.51	24.93
LineFieldOrientation	47.17	98.67	+51.50	98.64
ShapeColorObjects	19.58	61.58	+42.00	62.64
TextureMosaic	28.64	87.19	+58.55	88.73
Unweighted mean	29.39	56.75	+27.37	56.74
<i>Test datasets</i>				
CIFAR-100	21.80	38.87	+17.07	38.86
RESISC45	35.65	66.49	+30.84	65.97
Stanford Cars	4.45	7.29	+2.84	6.44
SUN397 (partition 1)	6.16	17.76	+11.60	18.05
Unweighted mean	17.02	32.60	+15.59	32.33

Table 4: Frozen linear-probe accuracy (%) at 100 million pre-training images, with the zero-image checkpoint as the untrained reference. Both IRC endpoints use the same evaluation protocol. Gains compare the primary run with its own initialization and are computed before rounding.

At 20 million images, mean validation accuracy is 52.93% and mean test accuracy is 30.67%. The corresponding fixed final values are 56.75% and 32.60%. Most of the improvement occurs before 20 million images, although individual datasets do not improve monotonically.

On CIFAR-10, final accuracy is 63.06%, whereas Stanford Cars remains at 7.29%. The strong synthetic-task results therefore should not be interpreted as equally strong object semantics. Orientation, procedural texture, and simple shape discrimination are closer to the regularities of the generated images than fine-grained recognition of real objects. The gains on real datasets show transfer beyond these synthetic tasks.

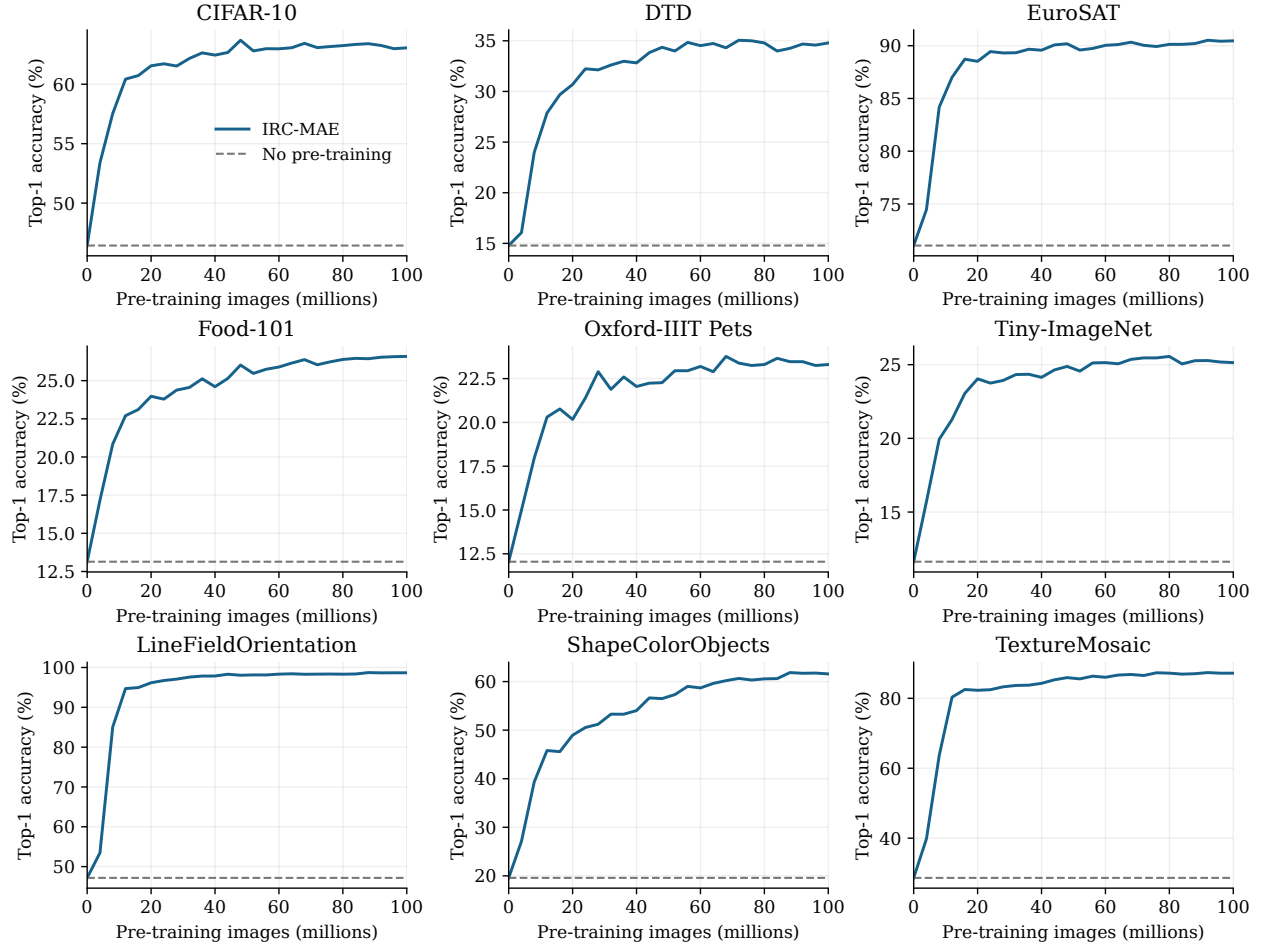


Figure 3: Frozen linear-probe accuracy on the nine validation datasets. The dashed reference uses the untrained encoder. Higher is better; the endpoint is fixed at 100 million images.

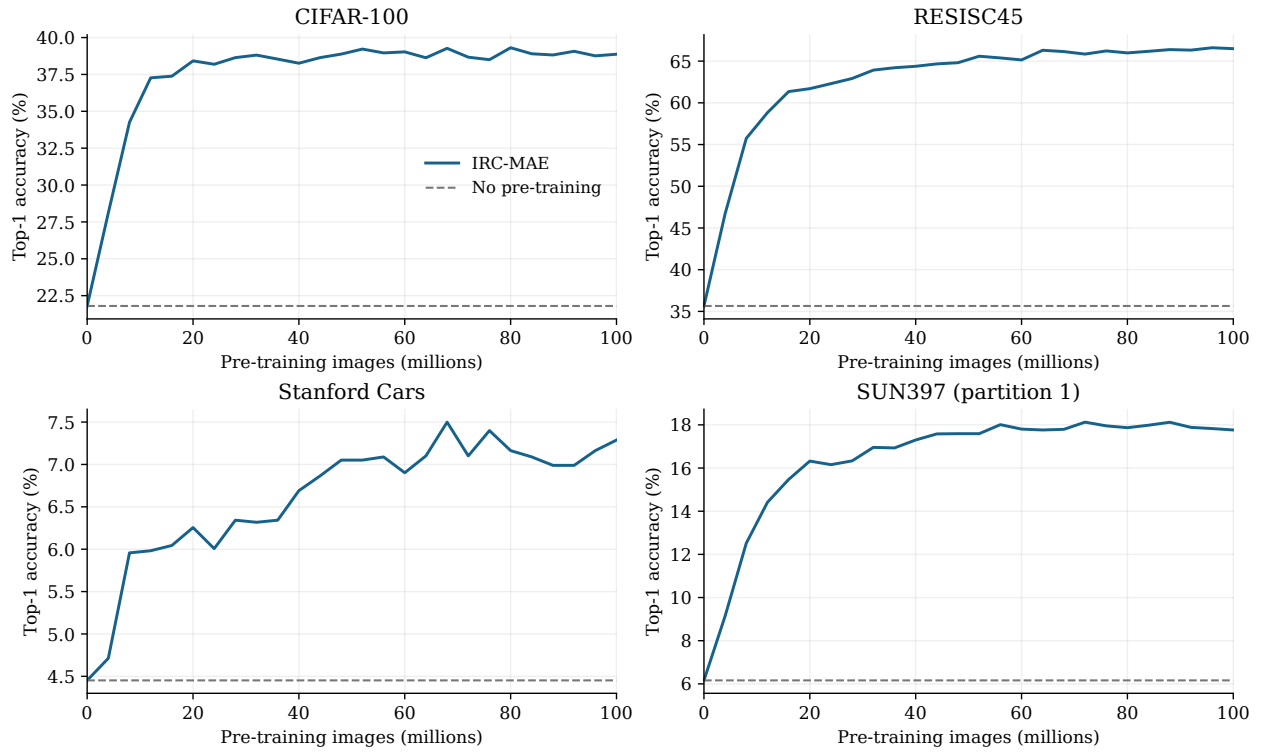


Figure 4: Frozen linear-probe accuracy on the four test datasets. The dashed reference uses the untrained encoder. Higher is better; the endpoint is fixed at 100 million images.

6.3 Reconstruction Without Adaptation

Figures 5 and 6 report masked raw-pixel MSE on each complete evaluation split. The curves begin at 4 million images and use linear axes, with the untrained error printed separately in each panel so that the much larger initial loss does not obscure differences between trained checkpoints. Because reconstruction is an average squared pixel error, a large reduction does not establish that the model has learned object semantics or preserves all high-frequency detail. The mean validation MSE changes from 0.54813 before pre-training to 0.01462 at the final checkpoint. The mean test MSE changes from 0.52377 before pre-training to 0.01665 at the final checkpoint.

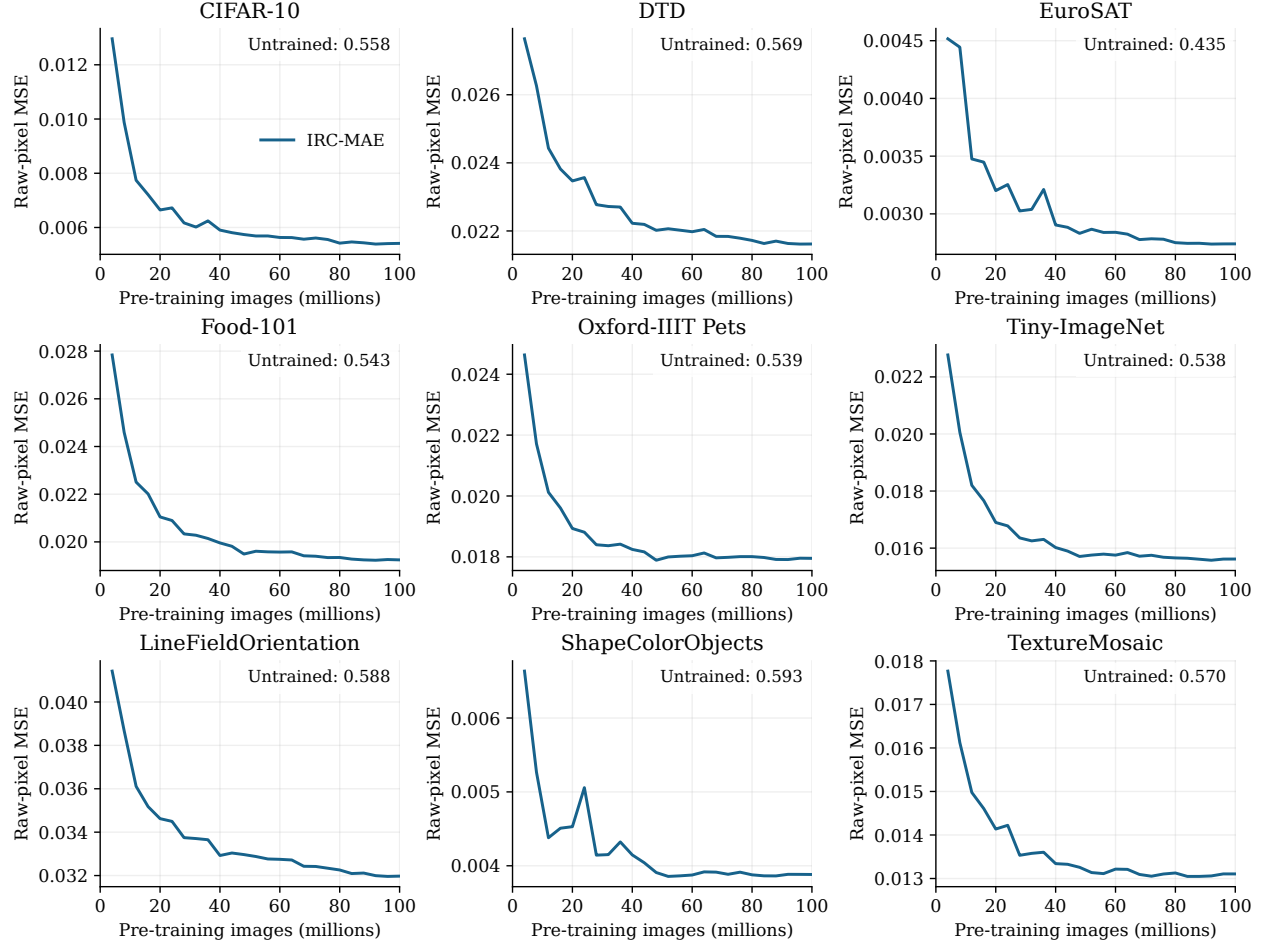


Figure 5: Raw-pixel reconstruction MSE on the nine validation datasets, without downstream adaptation. Curves show 4–100M images; the untrained loss is annotated separately. Lower is better. The mask ratio is 0.75 and patch normalization is disabled.

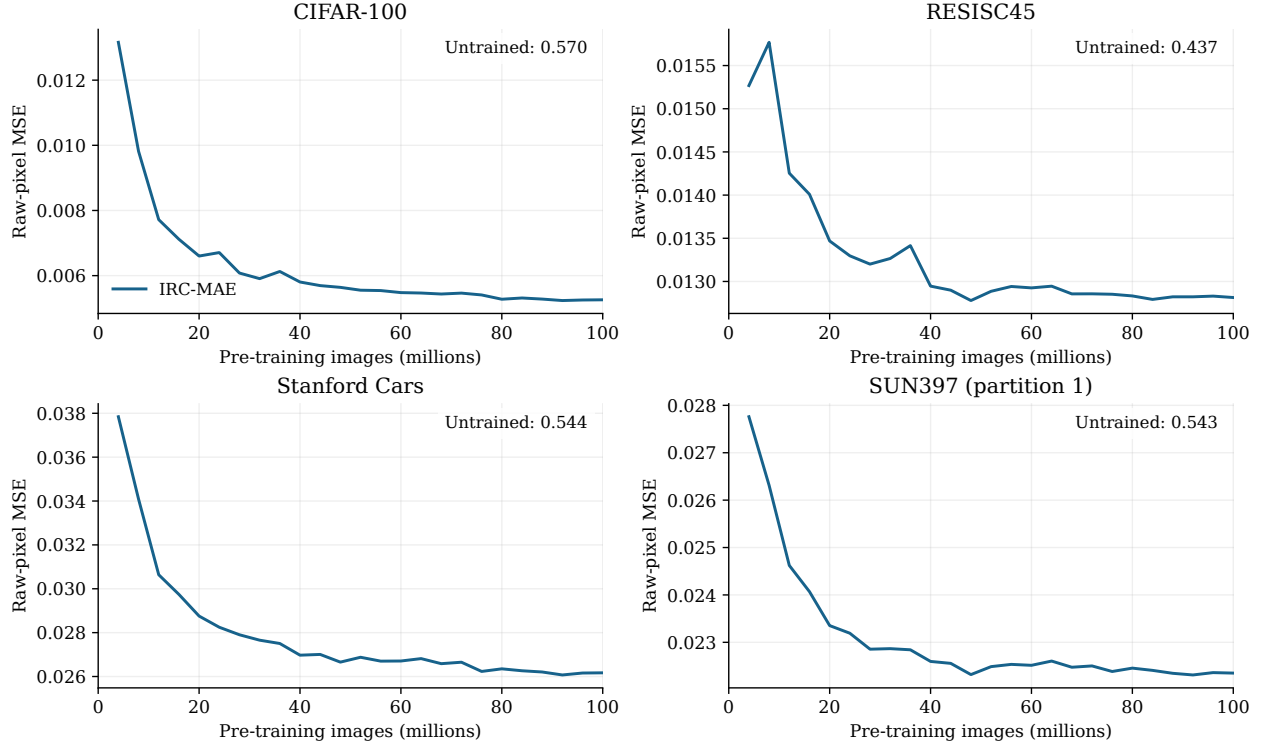


Figure 6: Raw-pixel reconstruction MSE on the four test datasets, without downstream adaptation. Curves show 4–100M images; the untrained loss is annotated separately. Lower is better. The mask ratio is 0.75 and patch normalization is disabled.

6.4 Historical Pre-Training Sources

Figure 7 and Table 5 use the original archived validation evaluations. All four runs were evaluated at 100 million images. These archived measurements are reported separately from the re-evaluated checkpoints because evaluator versions and numerical conditions may differ.

Historical source	Mean accuracy (%)	Mean raw MSE
IRC-MAE (historical)	56.71	0.01465
Fourier texture	40.51	0.01960
Video (continual)	42.74	0.01540
Video (stratified)	41.67	0.01603

Table 5: Archived 100 million-image results, averaged equally across the nine validation datasets. The source comparisons use the archived evaluation protocol.

The historical IRC run has a higher mean validation accuracy than the recorded Fourier and video runs. However, this does not show that it wins every dataset, and it does not isolate the effect of the persistent buffer, neural computations, or noise sources. Classification and reconstruction must also be distinguished: an image source can give a useful pixel-prediction task without yielding the strongest frozen classifier.

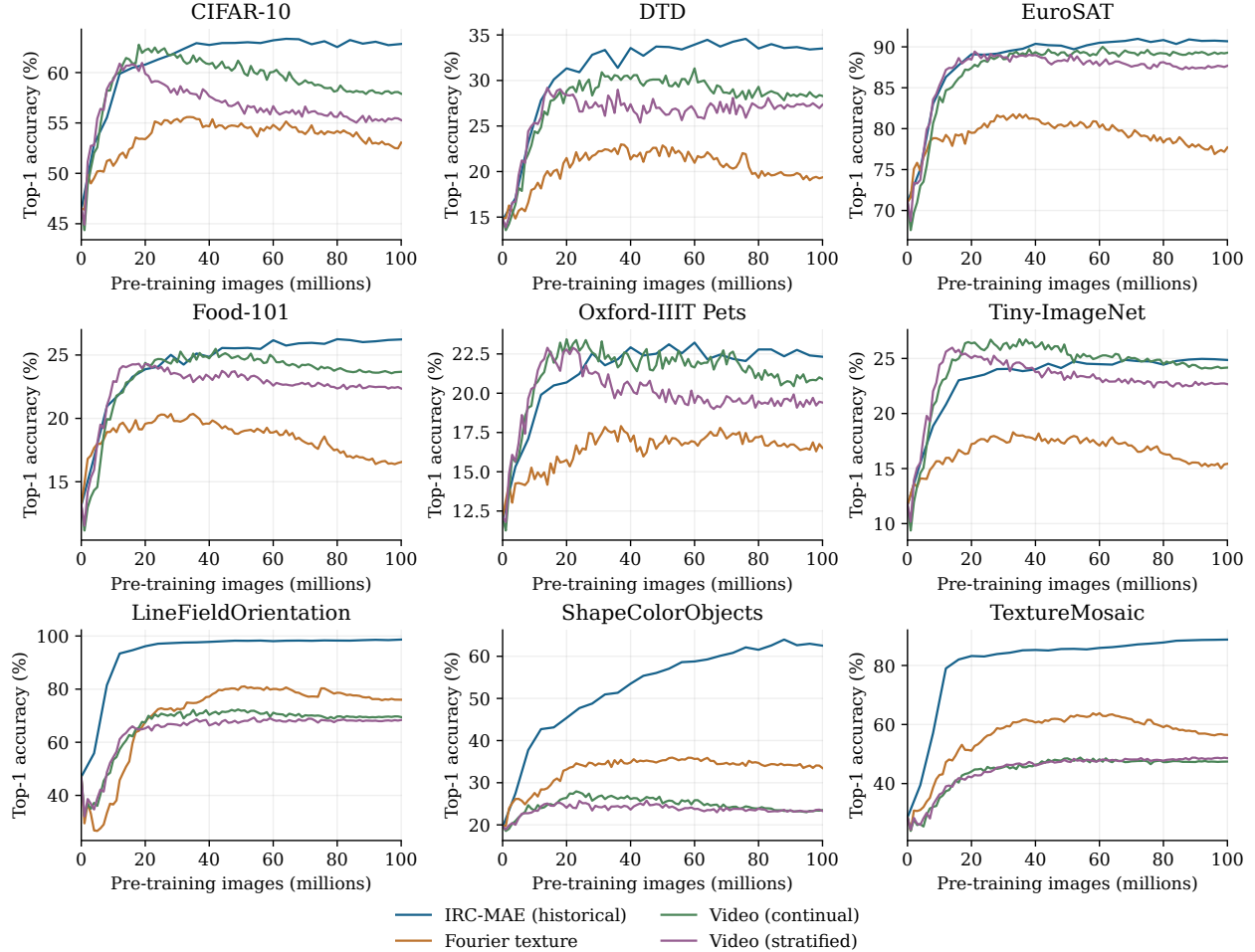


Figure 7: Archived validation curves for IRC-MAE, Fourier texture, and two real-video sampling variants. Each curve comes from a separate historical run.

6.5 End-to-End Fine-Tuning

Linear probing measures the usefulness of fixed representations. To test whether the initialization also helps when the encoder can adapt, we fine-tune the complete ViT-Tiny on two labeled subsets: ImageNet-100 using the CMC class selection, and ImageNet-1K using the SimCLR 10% training subset. These experiments use a separate 100 million-image IRC-MAE checkpoint; the primary run is used only for the frozen-probe and reconstruction evaluations. We compare this pre-trained initialization with the same architecture trained without pre-training.

Both conditions use 300 epochs, batch size 1,024, five warm-up epochs, and AdamW. The base learning rate is 2.5×10^{-4} , scaled by batch size over 256 to a peak of 10^{-3} , and decayed to 5×10^{-5} . Training uses random resized crops, horizontal flips, RandAugment, random erasing, Mixup/CutMix, and label smoothing. The comparison below uses layer decay 1.0 in both conditions and reports the final epoch rather than selecting the best checkpoint separately for each condition.

IRC-MAE gives a larger gain on the ImageNet-1K subset, suggesting greater benefit with fewer labeled examples per class. Pre-training may supply reusable features that are harder to learn from limited labeled data. However, class count and task difficulty also differ, so sample size is not isolated.

Dataset	No pre-training	IRC-MAE, 100M	Gain (pp)
ImageNet-100 (CMC)	84.34%	85.54%	+1.20
ImageNet-1K (10%)	49.29%	54.53%	+5.23

Table 6: Final-epoch fine-tuning accuracy after 300 epochs, with matched layer decay of 1.0. Each entry is one training run. Accuracy uses the corresponding ImageNet validation split; 10% refers only to the ImageNet-1K training subset.

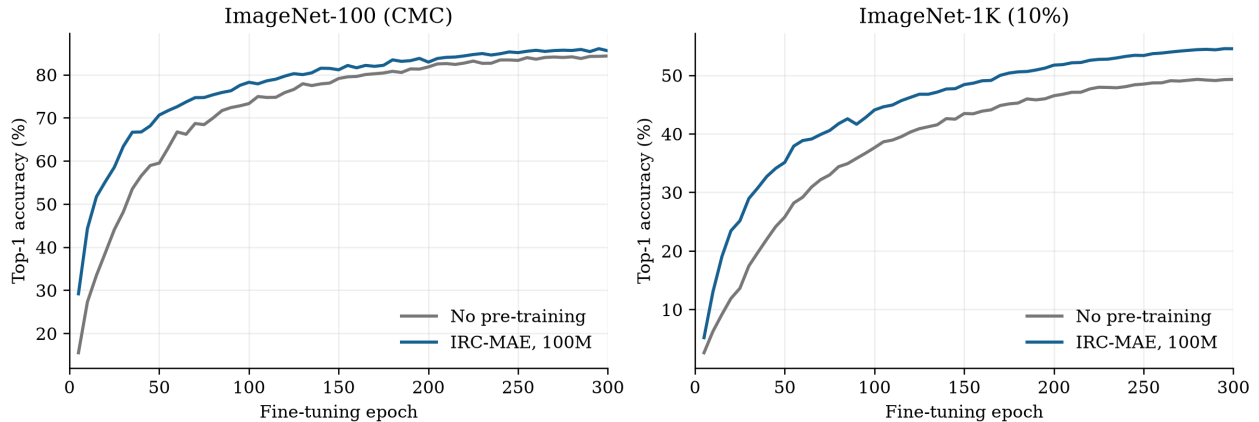


Figure 8: MAE fine-tuning accuracy throughout 300 epochs on the two ImageNet subsets. The no-pre-training and pre-trained conditions both use layer decay 1.0.

7 Limitations and Future Work

The experiments use a ViT-Tiny model and 100 million image presentations. Their results do not establish how the method scales to larger encoders or longer training budgets. Both model size and data scale affect visual pre-training [3, 12, 13]; repeating the configuration at larger scales would test whether the observed transfer gains persist.

The generator covers only a small part of the universal-distribution idea. The true universal distribution mixes over all computable data-generating processes. In contrast, this implementation uses small random residual convolutional networks, crop-and-resize, blur, crossover, a persistent buffer, and mixed noise/color resets. These choices favor local continuity, boundaries, color variation, and texture. They do not provide object labels or guarantee semantic composition. The recorded depth measures computational ancestry, but a higher depth need not correspond to greater visual complexity or greater learning difficulty. Blur can remove detail, and a crossover mask may copy almost entirely from one parent. A depth-conditioned learning experiment would be needed to test complexity directly.

The pre-training objective imposes another limitation. MAE reconstructs raw RGB patches [10], which can favor low-frequency pixel statistics over category-level representations. The I-JEPA experiment in Appendix A tests prediction in representation space, but the evaluated I-JEPA configuration has lower mean validation accuracy than MAE. The optimizer schedules, batch sizes, and encoder details also differ, so the comparison does not isolate the objective. Future work could test contrastive learning, self-distillation, or hybrid objectives while controlling the generator and optimization budget.

Each training condition uses one run, so the experiments do not estimate variability across random seeds. This is particularly important for the small I-JEPA fine-tuning gains. The fine-

tuning results come from separate checkpoints rather than a second evaluation of the primary model. Archived synthetic and video comparisons may also differ in implementation details. Prior use of the test datasets during the wider research project limits claims of evaluation on entirely unseen tasks.

Finally, the experiments do not isolate every design choice. Buffer persistence, noise initialization, crossover, blur, crop-and-resize, residual strength, reset rates, and depth-biased sampling may interact. Comparing the complete generator with a simpler source does not identify the contribution of any one operation. Controlled ablations would be needed before attributing the gain to random neural computation alone.

The improvement over an untrained encoder and the gains in the two MAE fine-tuning experiments show that synthetic computation can provide a useful visual initialization. Larger models and repeated evaluations are needed to determine how far these results generalize.

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A I-JEPA Experiments

A.1 I-JEPA Pre-Training

We also study I-JEPA [42] as an alternative to the masked-autoencoder objective. Instead of reconstructing RGB pixels, I-JEPA predicts the target encoder’s representations of masked image regions from a visible context. The target encoder is updated by an exponential moving average of the context encoder.

The target encoder processes the complete image. We apply layer normalization to its patch representations and then select the target-block positions. The predictor is trained with mean smooth-L1 loss against these representations. Gradients pass through the context encoder and predictor, while the target encoder is updated only through the EMA after each optimizer step.

The experiment uses 224×224 generated images and a ViT-Tiny/16 encoder with dimension 192, depth 12, and 3 attention heads. This encoder uses no class token and has fixed sinusoidal positional embeddings. The predictor has dimension 128, depth 6, and 4 heads. Four target blocks are sampled with area fractions between 0.15 and 0.20 and aspect ratios between 0.75 and 1.5. The nominal context area fraction lies between 0.85 and 1.0, with target locations excluded from the visible context. The target-encoder EMA coefficient increases from 0.992 towards 1.0.

The I-JEPA run processes 100.05 million images using the `irc_new` generator, with a global batch size of 4,002 and checkpoints every 4.002 million images. Training uses two distributed processes, each accumulating three microbatches of 667 images per optimizer step. AdamW uses $\beta = (0.9, 0.999)$, a 5% warm-up, a learning rate starting at 2×10^{-4} , peaking at 10^{-3} , and decaying to 10^{-6} , and weight decay increasing from 0.04 to 0.4. Evaluation uses the EMA target encoder and mean-pools its patch tokens before training the linear classifier. No pixel-reconstruction error is reported for I-JEPA.

At the final checkpoint, the archived evaluation gives a mean accuracy of 51.23% across the nine validation datasets, compared with 56.71% for the 100M IRC-MAE run. I-JEPA performs better on EuroSAT and Food-101 in these recorded runs, while MAE has the higher overall mean. This comparison concerns the tested configurations and single runs, rather than the general superiority of one self-supervised objective.

Dataset	No pre-training	IRC-I-JEPA, 100.05M	Gain (pp)
ImageNet-100 (CMC)	83.34%	83.62%	+0.28
ImageNet-1K (10%)	47.45%	47.79%	+0.34

Table 7: I-JEPA fine-tuning accuracy at epoch 300, with layer decay 1.0 in both conditions. The no-pre-training condition uses the corresponding I-JEPA encoder architecture and initialization.

Fine-tuning uses the same data splits, optimization settings, and augmentations as Section 6.5.

The fine-tuning gains are small on both datasets and require replication before drawing conclusions about their reliability.

A.2 Evaluation on Both Dataset Groups

The retained initial and final I-JEPA checkpoints are evaluated on the same 13 datasets as the primary MAE run. Table 8 uses the common re-evaluation protocol, while Figure 9 shows the original validation histories. Small differences from the archived endpoints arise from re-evaluation, not additional pre-training. The table compares the EMA target encoder before and after pre-training.

Dataset	I-JEPA untrained	I-JEPA, 100.05M	Gain (pp)
CIFAR-10	40.25	57.86	+17.61
DTD	9.95	29.95	+20.00
EuroSAT	56.06	91.30	+35.24
Food-101	7.72	29.48	+21.76
Oxford-IIIT Pets	10.38	20.47	+10.08
Tiny-ImageNet	7.05	24.50	+17.45
LineFieldOrientation	13.23	89.04	+75.81
ShapeColorObjects	14.01	38.24	+24.23
TextureMosaic	20.59	80.22	+59.63
Validation mean	19.91	51.23	+31.31
CIFAR-100	16.02	36.33	+20.31
RESISC45	24.79	69.10	+44.30
Stanford Cars	2.51	7.09	+4.58
SUN397 (partition 1)	4.33	15.71	+11.39
Test mean	11.91	32.06	+20.14

Table 8: Re-evaluated frozen I-JEPA accuracy (%) on both dataset groups, using the fixed final checkpoint.

The dataset composition matters when comparing objectives. Across the six real validation datasets, historical IRC-MAE and I-JEPA obtain mean accuracies of 43.43% and 42.26%, respectively. Across the three synthetic diagnostics, the corresponding means are 83.34% and 69.17%. Thus, much of the difference in the nine-dataset mean comes from synthetic-task transfer.

On the four test datasets, the means are much closer: 32.33% for historical IRC-MAE and 32.06% for I-JEPA. I-JEPA performs better on RESISC45 and Stanford Cars in these retained checkpoints, while MAE performs better on CIFAR-100 and SUN397. This mixed ordering supports comparing the objectives by task rather than treating the validation mean as a universal ranking. The separate fine-tuning results still show larger gains from MAE initialization under the tested settings.

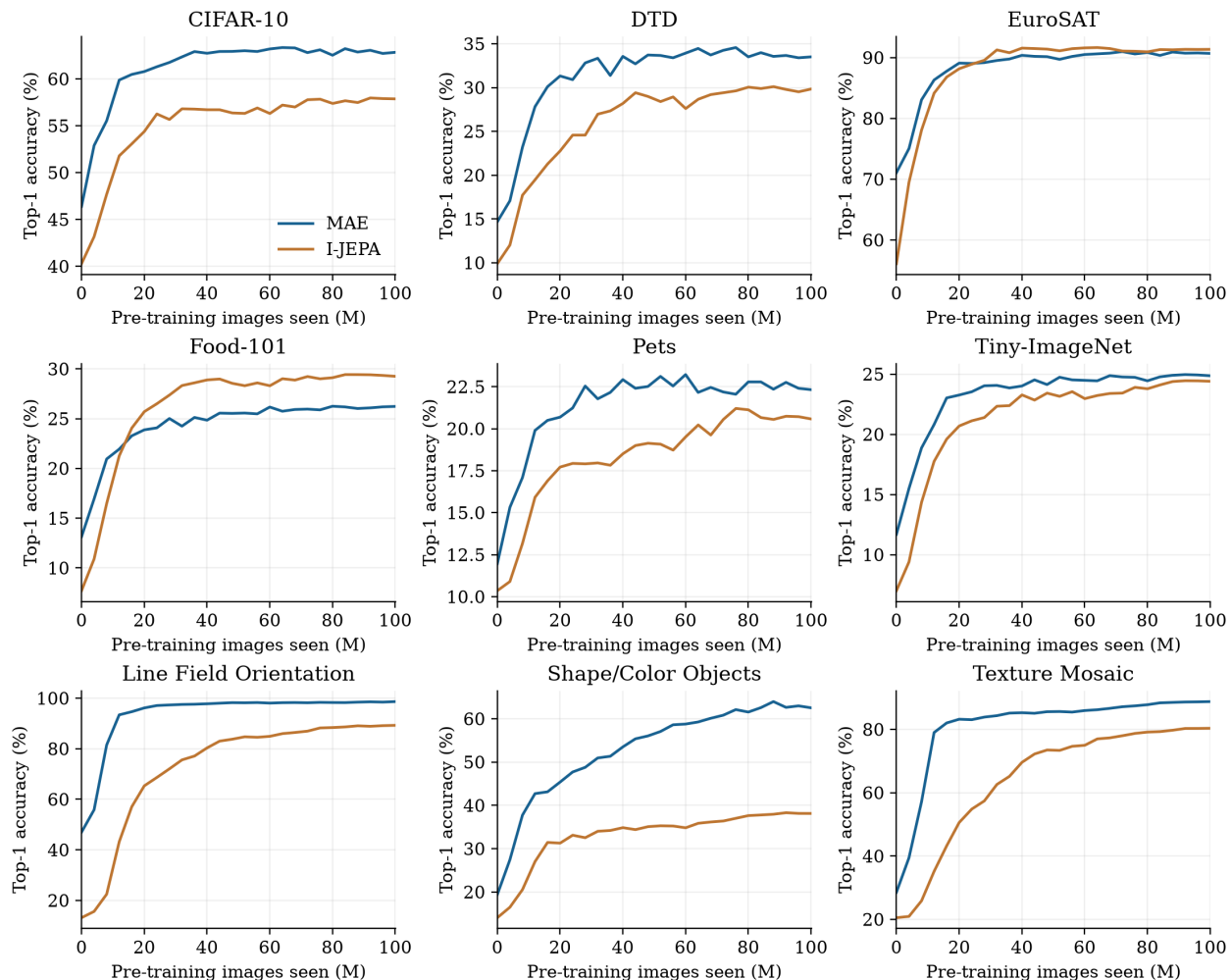


Figure 9: Original validation histories of the 100M IRC-MAE and 100.05M IRC-I-JEPA runs. The objectives and training schedules differ, so this comparison concerns the tested configurations.

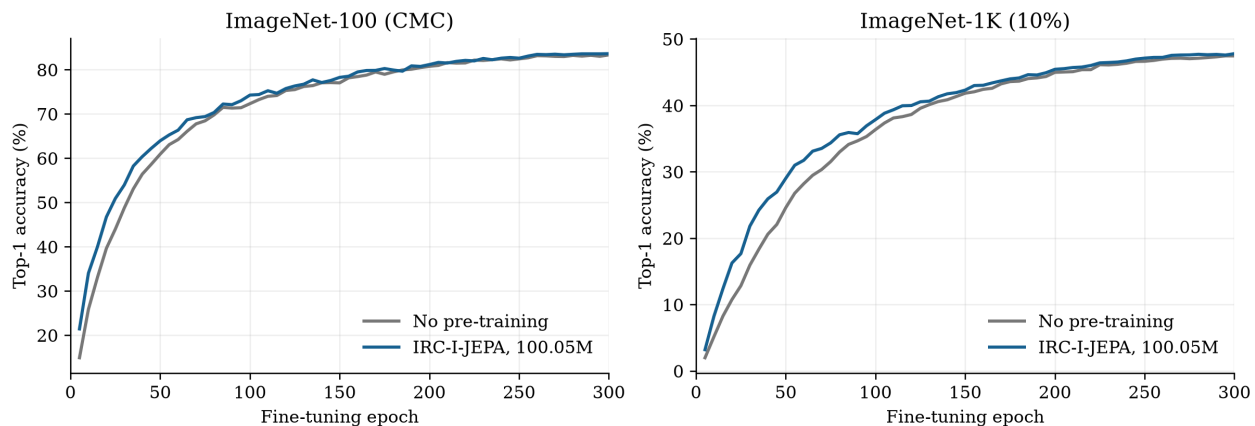


Figure 10: I-JEPA fine-tuning on the two ImageNet subsets. Both conditions use 300 epochs and layer decay 1.0.